

Comprehensive Cancer Center

Cancer Genetics Program

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Date of Service: December 18, 2024

Date of Letter: December 19, 2024

Patient Name: Jenny M

Date of Birth: January 1, 1965 (Age 55)

Medical Record Number: MRN-8847261

Referring Physician: Dr. Sandra Caldwell, MD, Oncology Associates

Dear Ms. M,

Thank you for attending your genetic counseling appointment at the Comprehensive Cancer Center on December 18, 2024. This letter summarizes the information discussed during your visit, including your personal and family history of cancer, hereditary cancer risk assessment, and recommendations for genetic testing.

Indication for Genetic Counseling

You were referred to our Cancer Genetics Program for evaluation of hereditary breast cancer risk following your recent diagnosis of breast cancer at age 55. Your medical history includes a diagnosis of primary left breast cancer, for which you have undergone a partial mastectomy (lumpectomy), biopsy with ultrasound guidance, and chemotherapy with cyclophosphamide, doxorubicin, and paclitaxel. You are currently on anastrozole for adjuvant hormonal therapy. In addition to your personal history, your family history of cancer warrants further assessment for hereditary cancer syndromes.

Personal Medical History

You reported being in generally good health prior to your breast cancer diagnosis. You have no history of ovarian, pancreatic, or other cancers. You underwent routine mammographic screening and had no prior abnormal findings until this year. Your menarche was at age 13, and you experienced menopause at age 51. You have two biological children, with first pregnancy at age 28. You have no history of hormone replacement therapy. You are a non-smoker and consume alcohol occasionally.

Family History Review

A detailed three-generation pedigree was constructed during your visit. The following family history of cancer was documented:

Maternal Side:

- **Mother:** Living, age 78, no history of cancer.
- **Maternal Aunt (mother's sister):** Diagnosed with breast cancer at age 48; treated with mastectomy and chemotherapy. She is currently 72 years old and without evidence of recurrence.
- **Maternal Uncle:** Living, age 75, history of colon polyps but no cancer diagnosis.
- **Maternal Grandfather:** Deceased at age 81 from cardiovascular disease; no known cancer history.
- **Maternal Grandmother:** Deceased at age 69 from complications of diabetes; no documented cancer history.

Paternal Side:

- **Father:** Deceased at age 70 from myocardial infarction; no history of cancer.
- **Paternal Aunt:** Deceased at age 66 from lung cancer (smoker).
- **Paternal Uncle:** Living, age 73, prostate cancer diagnosed at age 68, treated with radiation.
- **Paternal Grandparents:** Both deceased; paternal grandfather from stroke at age 76, paternal grandmother from pneumonia at age 79. No known cancer history.

Siblings and Children:

- **Sister:** Age 52, no history of cancer.
- **Brother:** Age 50, no history of cancer.
- **Daughter:** Age 27, healthy, no medical concerns.
- **Son:** Age 25, healthy, no medical concerns.

Your family history is significant for breast cancer in a first-degree relative on your maternal side (your aunt diagnosed at age 48), in addition to your own diagnosis at age 55. The paternal history of prostate cancer and lung cancer (likely environmental) is also noted.

Risk Assessment

Based on your personal history of breast cancer at age 55 and your family history of breast cancer in a maternal aunt diagnosed before age 50, your clinical presentation meets National Comprehensive Cancer Network (NCCN) criteria for hereditary breast and ovarian cancer (HBOC) genetic testing. Specifically, the following NCCN criteria apply:

- Personal history of breast cancer diagnosed at age ≤ 50 , or
- Personal history of breast cancer at any age with a first- or second-degree relative with breast cancer diagnosed at age ≤ 50 .

Hereditary breast and ovarian cancer syndrome is most commonly caused by pathogenic variants in the *BRCA1* and *BRCA2* genes, which account for approximately 5–10% of all breast cancers. Other moderate-

and high-penetrance genes associated with hereditary breast cancer include *PALB2*, *CHEK2*, *ATM*, *TP53*, *PTEN*, *CDH1*, and *STK11*.

Identification of a pathogenic variant has important implications for:

- Your ongoing cancer surveillance and management, including consideration of risk-reducing strategies;
- Your family members, who may be at increased risk and eligible for predictive testing;
- Potential eligibility for targeted therapies such as PARP inhibitors.

Genetic Testing Recommendation

After discussion of the benefits, limitations, and potential risks of genetic testing, you elected to proceed with multi-gene panel testing for hereditary cancer syndromes. The recommended panel includes analysis of the following genes:

- *BRCA1* and *BRCA2*
- *PALB2*
- *CHEK2*
- *ATM*
- *TP53*
- *PTEN*
- *CDH1*
- *STK11*
- *NBN*
- *BARD1*
- *RAD51C* and *RAD51D*

Testing will be performed using next-generation sequencing to detect pathogenic and likely pathogenic variants, as well as large deletions and duplications.

Informed Consent Discussion

We reviewed the elements of informed consent for genetic testing, including:

- **Possible Results:** Testing may identify a pathogenic variant (positive), no pathogenic variant (negative), or a variant of uncertain significance (VUS). A VUS is a change in the gene for which there is currently insufficient evidence to classify it as harmful or benign.
- **Implications of Results:** A positive result confirms hereditary cancer predisposition and informs medical management and family testing. A negative result does not eliminate all hereditary risk but significantly lowers the probability. A VUS does not change medical management.

- **Insurance and Discrimination:** The Genetic Information Nondiscrimination Act (GINA) provides federal protection against genetic discrimination in health insurance and employment. GINA does not apply to life, disability, or long-term care insurance.
- **Psychosocial Impact:** Genetic test results can have emotional and psychological implications for you and your family. Our team is available to provide support and counseling.
- **Privacy and Confidentiality:** Your genetic information is protected under HIPAA. Results will be disclosed only to individuals you authorize.

You expressed understanding of these points and provided written informed consent for testing.

Sample Collection and Next Steps

A blood sample was collected at the conclusion of today's visit and will be sent to GeneDx Laboratories for analysis. Genetic test results are typically available within 3 to 4 weeks. Once results are received, we will contact you to schedule a follow-up appointment to review the findings and discuss their implications for your medical care and family.

In the interim, please continue with your current oncology care plan as directed by Dr. Caldwell. If you have any questions or concerns prior to your results appointment, please do not hesitate to contact our office at (217) 555-8820.

Summary and Recommendations

- Personal history of left breast cancer at age 55.
- Family history significant for breast cancer in a maternal aunt at age 48.
- Meets NCCN criteria for hereditary breast and ovarian cancer syndrome genetic testing.
- Multi-gene panel testing ordered; sample collected on December 18, 2024.
- Results expected in approximately 3 weeks; follow-up appointment to be scheduled.
- Continue routine oncology care and breast cancer treatment as planned.

It was a pleasure meeting with you, Ms. M. Please feel free to reach out if you have additional questions or if there are changes to your family history that should be documented. We look forward to reviewing your results with you soon.

Sincerely,

Laura K. Brennan, MS, LCGC

Licensed Certified Genetic Counselor

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